The near-efficient market of Grossman and Stiglitz fits closely with the multiple factor risk framework of the APT developed by Ross (1976). In Ross's multifactor model, active managers and arbitrageurs drive the expected return of assets toward a value consistent with an equilibrium trade-off between risk and return. The factors in the APT model are systematic ones, or those that affect the whole economy, that agents wish to hedge against. In their purest form the factors represent risk that cannot be arbitraged away, and investors need to be compensated for bearing this risk.

Despite the modern notion that markets are not perfectly efficient, a large literature continues to test the *Efficient Market Hypothesis* (EMH). The implication of the EMH is that, to the extent that speculative trading is costly, active management is a loser's game and investors cannot beat the market.[14] The EMH does give us a very high benchmark: if we are average, we hold the market portfolio and indeed we come out ahead simply because we save on transactions costs. Even if we know the market cannot be perfectly efficient, tests of the EMH are still important because they allow investors to gauge where they may make excess returns. In the Grossman–Stiglitz context, talented investors can identify the pockets of inefficiency where active management efforts are best directed.

The EMH has been refined over the past several decades to rectify many of the original shortcomings of the CAPM including imperfect information and the costs associated with transactions, financing, and agency. Many behavioral biases have the same effect and some frictions are actually modeled as behavioral biases. A summary of EMH tests is given in Ang, Goetzmann, and Schaefer (2011). What is relevant for our discussion is that the deviations from efficiency have two forms: rational and behavioral. For an asset owner, deciding which prevails is important for deciding whether to invest in a particular pocket of inefficiency.

In a rational explanation, high returns compensate for losses during bad times. This is the pricing kernel approach to asset pricing. The key is defining those bad times and deciding whether these are actually bad times for an individual investor. Certain investors, for example, benefit from low economic growth even while the majority of investors find these to be bad periods. In a rational explanation, these risks premiums will not go away—unless there is a total regime change of the entire economy. (These are very rare, and the financial crisis in 2008 and 2009 was certainly not a regime change.) In addition, these risk premiums are scalable and suitable for very large asset owners.

In a behavioral explanation, high expected returns result from agents' under- or overreaction to news or events. Behavioral biases can also result from the inefficient updating of beliefs or ignoring some information. Perfectly rational investors, who are immune from these biases, should be able to come in with sufficient capital and remove this mispricing over time. Then it becomes a question of

[14] See Ellis (1975) for a practitioner perspective.